BASANT KUMAR

AI & Business Analyst | Business Analytics · Data · AI
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Portfolio · LinkedIn · GitHub

PROFILE

Business analytics graduate with an electronics engineering background, working across business problems, data and AI. I completed the MSc in Business Analytics at Maynooth University in 2026 (award pending), after R&D work in IoT and analytics and research reporting at Maynooth. I like framing the problem with stakeholders first, getting the data into a state worth trusting, modelling it, and turning the result into something people can use. Seeking Business Analyst, Data Analyst, BI Analyst and AI/Product Analyst roles in Ireland.

EDUCATION

MSc Business Analytics, Maynooth University, Ireland 2024 – 2026
Programme completed 2026; award pending

BEng Electronics & Telecommunication Engineering Aug 2016 – Apr 2020
Savitribai Phule Pune University, India

PROFESSIONAL EXPERIENCE

Graduate Research Assistant, Maynooth University Nov 2024 – Feb 2026
Kildare, Ireland

- Built automated reporting dashboards using Python, R and Tableau.
- Interviewed HR leads, CEOs and senior managers, then analysed the qualitative data to identify themes.

R&D Engineer, IoT & Analytics, Bajaj Electricals Ltd Oct 2022 – Dec 2023
India

- Analysed IoT sensor data, built KPI reporting and applied forecasting.
- Worked with cross-functional teams on requirements and technical deliverables.

Junior Engineer, AI & IoT Solutions, Bajaj Electricals Ltd Jul 2021 – Oct 2022
India

- Analysed existing processes and prepared gap-analysis reports.
- Built KPI dashboards and supported performance benchmarking across AI and IoT solutions.

E-commerce Founder (independent business) Feb 2018 – Aug 2019
Delhi, India

- Ran a Shopify business across sales, inventory, suppliers and marketing.
- Used commercial data to understand demand, conversion and stock requirements.

SELECTED PROJECTS

BeBeyond: AI fitness and nutrition platform Personal product
Private build, in development · React Native, Expo, TypeScript, FastAPI, PostgreSQL, Three.js

- Designed and built end to end: exercise ontology, muscle load and readiness with confidence tiers, recommendation and nutrition engines, and a 3D exercise view.
- Deterministic engines own every number; a rule-based coach explains each recommendation. Language-model explanations are written but switched off by default.
- Camera form analysis runs on a simulated pose source; wearable adapters are scaffolded with no live data.

Fraud Detection: explainable XGBoost model MSc team project
Coursework MI6228, Maynooth University · Python, XGBoost, GridSearchCV, SHAP, scikit-learn, Gradio

- **My contribution:** feature engineering, XGBoost tuning, SHAP analysis and Gradio deployment.

- Engineered behavioural features (spend deviation from the cardholder average, distance to merchant, card-not-present proxy) and tuned XGBoost with a precision-first threshold.
- On a 166,716-transaction hold-out from 555,719 synthetic Kaggle transactions, the tuned model caught 496 of 644 frauds versus 59 for a logistic regression baseline, with 11 false alerts; AUC-ROC 99.9%.

Housing Accessibility: client project for Age Friendly Ireland MSc team project
Four-person consultancy team · CSO Census 2022, CSO PEC19 population projections

- Applied Census 2022 prevalence by age and sex to population projections and converted it to household level: an estimated 16.9% of households in 2022, projected to reach 22.3% by 2050 (100,000+ additional households).
- Benchmarked housing-adaptation policy in Australia, Sweden and Denmark; five recommendations delivered to the client in an executive presentation.

Sales Demand Forecasting: ARIMA vs Prophet
R, forecast, prophet, dplyr, ggplot2

- Compared Auto ARIMA and Prophet on product sales data with rolling-window validation on RMSE, MAE and MAPE.
- Prophet performed better on strongly seasonal series with holiday effects; ARIMA on stationary, trend-driven series.

Swarm Drone Coordination System 2019 – 2020
Bachelor's final-year team project · Arduino, GPS, gyroscope, RF modules

- **Achievement (2018):** Qualified in the DST & Texas Instruments India Innovation Challenge Design Contest 2018, anchored by IIM Bangalore.
- Developed a two-drone coordination prototype for disaster-response applications, exploring coordinated operations for supply delivery and search-and-rescue support.
- Integrated Arduino-based control hardware with GPS, gyroscope and RF modules as part of the engineering system.
- Explored navigation, drone coordination and operational safety considerations for a practical disaster-response application.

SKILLS

Business analysis: requirements, process improvement, stakeholder management, KPI definition, As-Is / To-Be, gap analysis, supply chain analytics, project management
Data & analytics: Python (analysis, feature engineering, ML), SQL (querying, data extraction), R (statistics, visualisation, forecasting), Tableau (dashboards, reporting), Power BI, Excel, Power Query, MySQL
Machine learning: XGBoost, SHAP, scikit-learn, model evaluation, forecasting (ARIMA, Prophet)
Product & engineering: FastAPI, PostgreSQL, React Native, TypeScript, APIs, IoT, Git and GitHub
Engineering tools: MATLAB, CAD

CERTIFICATIONS

Google Data Analytics Professional Certificate Google, 2024
Google Project Management Professional Certificate (six courses) Google, 2022
Machine Learning Professional Altair / RapidMiner, 2025
Courses: AI For Everyone, DeepLearning.AI (2023) · Programming for Everybody, University of Michigan (2019) · Six Sigma Principles, University System of Georgia (2023) · Supply Chain Management Strategy, Rutgers (2022) · Power BI, be10x one-day workshop (2023)

LEADERSHIP & VOLUNTEERING

- **Save the Children, Pune:** weekend volunteer raising awareness of girls' education, teaching mathematics to children and fundraising in the community.
- **Head Boy** at higher secondary school; **Fashion Show Head** at college, mentoring team members competing at events around Pune; college cricket team.